

Invent Cast

Deep Intelligence Report

<https://inventcast.in/>

Executive Summary

Invent Cast is a prominent player in the industrial manufacturing sector, specializing in the production of precision-engineered components through advanced investment casting and machining solutions. The company offers a diverse range of high-quality nickel and nickel alloy castings, designed to meet the stringent demands of various industrial applications. With a focus on high dimensional accuracy and the capability to produce complex geometries with repeatable precision, Invent Cast positions itself as a reliable partner for manufacturing and production managers seeking cost-effective solutions for low- to medium-volume requirements.

Founded with a commitment to excellence, Invent Cast has established itself as a leader in the market by leveraging an integrated manufacturing facility equipped with advanced quality assurance processes. This commitment to quality ensures compliance with international standards, including ASTM, and allows the company to deliver products that exhibit excellent mechanical properties and versatility in size and material flexibility. By addressing the specific needs of its target audience across multiple geographies, including the United States, UK, India, and several others, Invent Cast differentiates itself through its precision-driven approach and comprehensive understanding of the manufacturing landscape, making it a preferred choice for businesses looking for reliable and high-quality industrial components.

Business Identity

Official Name: Invent Cast

Industry: Manufacturing

Sub-Industry: Industrial manufacturing

Business Model: B2B manufacturing

Product Catalog

Investment Castings

Category: Investment Castings

Precision-engineered components manufactured using investment casting techniques.

Key Features:

- High dimensional accuracy
- Complex geometries
- Repeatable accuracy

Technical Specifications:

tensileStrength	Varies by material
yieldStrength	Varies by material
elongation	Varies by material

Use Cases:

- Used in the manufacturing of valves and pumps where precision and durability are critical.
- Applied in automotive components requiring high strength and corrosion resistance.

Export Markets:

- Germany
- USA
- Europe

Investment Castings Summary:

Investment Castings are specialized parts made through a precise manufacturing process called investment casting. This method allows for the creation of components that are not only very accurate in size and shape but can also feature complex designs that would be difficult to achieve with other manufacturing techniques. Essentially, it involves pouring molten metal into a mold created from a wax pattern, which is then melted away, leaving a hollow cavity that is filled with the metal. This results in parts that are highly detailed and consistent in quality.

Benefits of Investment Castings:

- **High Dimensional Accuracy:** The process ensures that each piece is made to exact specifications, which is crucial for parts that need to fit together perfectly.
- **Complex Geometries:** Manufacturers can create intricate designs that are lightweight yet strong, making them ideal for advanced applications.
- **Repeatable Accuracy:** Once the mold is made, it can produce many identical parts, making it efficient for large-scale production.

Investment Castings are widely used in industries such as automotive and manufacturing, particularly for components like valves and pumps where durability and precision are essential. These castings are exported to markets like Germany, the USA, and throughout Europe, showcasing their global demand.

ASTM A 494 N12MV Castings

Category: Investment Castings

Nickel and Nickel Alloy Castings intended for various industrial applications.

Key Features:

- High corrosion resistance
- Suitable for high-temperature applications

Technical Specifications:

Tensile Strength	525 Mpa min
Yield Strength	275 Mpa min
Elongation	6% min

Use Cases:

- Used in valve and pump components where high corrosion resistance is required.

ASTM A 494 N12MV Castings Summary:

ASTM A 494 N12MV Castings are specialized metal castings made from nickel and nickel alloys, designed for use in demanding industrial environments. These castings are particularly useful in applications where components need to withstand harsh conditions, especially in terms of corrosion and high temperatures.

Key benefits include:

- **High Corrosion Resistance:** This means that the castings can resist damage from chemicals and moisture, making them ideal for use in valves and pumps that are exposed to corrosive substances.
- **High-Temperature Suitability:** These castings can perform well even in extreme heat, ensuring reliability and longevity in applications where high temperatures are common.

Overall, ASTM A 494 N12MV Castings provide durable and reliable solutions for industries that require high-performance materials for their equipment.

ASTM A 511 N08904 Seamless Stainless Steel Mechanical Tubing

Category: Investment Castings

Seamless stainless steel tubing for mechanical applications.

Key Features:

- High tensile strength
- Excellent corrosion resistance

Technical Specifications:

Tensile Strength	490 Mpa min
Yield Strength	215 Mpa min
Elongation	35% min

Use Cases:

- Ideal for use in automotive and general engineering applications requiring high strength and corrosion resistance.

ASTM A 511 N08904 Seamless Stainless Steel Mechanical Tubing Summary:

ASTM A 511 N08904 is a type of seamless stainless steel tubing specifically designed for mechanical uses. This tubing is known for its impressive strength and ability to withstand corrosion, making it an excellent choice for various applications. Its seamless design means there are no welds or joints, which enhances its durability and reliability in tough conditions.

- Key benefits of ASTM A 511 N08904 include:
- **High Tensile Strength:** With a minimum tensile strength of 490 Mpa, it can handle heavy loads without breaking.
 - **Corrosion Resistance:** This tubing resists rust and other forms of corrosion, ensuring longevity in challenging environments.

This product is particularly well-suited for automotive and engineering projects where strength and resistance to harsh conditions are crucial.

ASTM A 580 S31277 Stainless Steel Wire

Category: Investment Castings

Stainless steel wire for industrial applications.

- Key Features:**
- High tensile strength
 - Corrosion resistant

Technical Specifications:

Tensile Strength	1620 Mpa min
Yield Strength	1345 Mpa min

- Use Cases:**
- Used in automotive and engineering applications where high strength and corrosion resistance are critical.

ASTM A 580 S31277 Stainless Steel Wire Summary:

ASTM A 580 S31277 Stainless Steel Wire is a strong and durable wire designed for industrial use, particularly in automotive and engineering sectors. It is made from high-quality stainless steel, which ensures that it can withstand harsh conditions without rusting or breaking. This wire is essential for applications where both strength and resistance to corrosion are crucial, making it a reliable choice for manufacturers.

- Key Benefits:**
- **High Tensile Strength:** It can handle heavy loads without snapping, ensuring safety and reliability in various applications.
 - **Corrosion Resistant:** It won't rust, even in damp or harsh environments, prolonging the lifespan of products made with it.

Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes

Category: Investment Castings

Tubes made from seamless ferritic and austenitic alloy-steel, intended for use in boilers, superheaters, and heat exchangers.

- Key Features:**
- High corrosion resistance
 - Suitable for high-temperature applications

Technical Specifications:

Tensile Strength in Mpa	490 min
Yield Strength in Mpa	215 min
Elongation in %	35 min
Hardness in HRC	90 max

Use Cases:

- Used in the construction of boilers and heat exchangers where high temperature and corrosion resistance are required.

Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes Summary:

The Seamless Ferritic and Austenitic Alloy-Steel Tubes are specially designed pipes that are used in high-temperature equipment like boilers and heat exchangers. They are made from a strong type of steel that is both seamless and resistant to rust and corrosion, which makes them ideal for tough industrial environments.

Benefits of These Tubes:

- **High Strength:** They can withstand significant pressure and temperature, ensuring safety and efficiency in operations.
- **Corrosion Resistance:** Their unique material helps prevent damage from rust, extending the lifespan of the equipment they are used in.
- **Versatile Use:** Perfect for any application that requires high performance in challenging conditions, making them essential for energy and manufacturing industries.

Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes

Category: Investment Castings

Tubes made from austenitic steel, designed for use in boilers, superheaters, heat exchangers, and condensers.

Key Features:

- High tensile strength
- Excellent corrosion resistance

Technical Specifications:

Tensile Strength in Mpa	770 min
Yield Strength in Mpa	360 min
Elongation in %	40 min
Hardness in HRC	100 max

Use Cases:

- Ideal for use in high-pressure and high-temperature environments such as power plants and chemical processing facilities.

Austenitic Steel Boiler, Superheater, Heat-Exchanger, and Condenser Tubes Summary:

Austenitic Steel Tubes are specialized tubes made from a strong and durable type of steel, specifically designed for use in various industrial applications like boilers, superheaters, heat exchangers, and condensers. These tubes are built to withstand high pressure and temperature, making them perfect for power plants and chemical processing facilities.

Key Benefits:

- **High Strength:** They can handle significant stress without breaking, ensuring reliable performance.
- **Corrosion Resistance:** They resist rust and damage from harsh chemicals, which prolongs their lifespan and reduces maintenance costs.

Castings, Iron-Chromium, Iron-Chromium-Nickel Corrosion Resistant

Category: Investment Castings

Corrosion-resistant castings made from iron-chromium and iron-chromium-nickel alloys.

Key Features:

- Corrosion resistance
- Suitable for general engineering applications

Technical Specifications:

Tensile Strength in Mpa	450 min
Elongation in %	4 min

Use Cases:

- Used in the manufacturing of components for valves and pumps where corrosion resistance is critical.

Castings, Iron-Chromium, Iron-Chromium-Nickel Corrosion Resistant Summary:

Castings made from iron-chromium and iron-chromium-nickel alloys are specialized metal pieces designed to resist rust and corrosion. These materials are particularly useful in making parts for machinery like valves and pumps, where exposure to harsh environments could cause damage. Their robust nature ensures that they can withstand challenging conditions while maintaining their structural integrity.

Key Benefits:

- **Corrosion Resistance:** These castings are built to last, even in environments where other materials might fail.
- **Versatility:** They are suitable for various engineering applications, making them a reliable choice for manufacturers.
- **Strong Performance:** With a minimum tensile strength of 450 MPa and a minimum elongation of 4%, they provide durability and flexibility in use.

ASTM A 946 Grade S70003

Category: Plate, Sheet and Strip

Chromium, Chromium-Nickel and Silicon Alloy Steel for Corrosion and Heat Resisting Service

Key Features:

- Corrosion and heat resistance
- Suitable for various industries

Technical Specifications:

Tensile Strength	540 Mpa min.
Yield Strength	240 Mpa min.
Elongation	50% min.
Hardness	217 BHN max.

Use Cases:

- Used in manufacturing components for valves and pumps due to its corrosion and heat resistance.

ASTM A 946 Grade S70003 Summary:

ASTM A 946 Grade S70003 is a special type of steel that is designed to resist rust and high temperatures, making it ideal for tough environments. It is commonly used in various industries to create strong and durable components, particularly in the manufacturing of valves and pumps. This steel's unique composition includes chromium, nickel, and silicon, which enhance its ability to withstand challenging conditions.

- Key benefits of ASTM A 946 Grade S70003 include:
- Excellent resistance to corrosion and heat, ensuring long-lasting performance.
 - Versatile application across different industries, making it a reliable choice for manufacturing important machinery components.
 - Strong structural characteristics, with a tensile strength of at least 540 Mpa and good elongation properties, allowing it to bend and stretch without breaking.

ASTM A 953 S70003

Category: Tube

Austenitic Chromium-Nickel-Silicon Alloy Steel Seamless Tubing

- Key Features:**
- High corrosion resistance
 - Suitable for high-temperature applications

Technical Specifications:

Tensile Strength	540 Mpa min.
Yield Strength	240 Mpa min.
Elongation	40% min.
Hardness	95 HRC max.

- Use Cases:**
- Ideal for use in high-temperature and corrosive environments in valves and pumps.

ASTM A 953 S70003 Summary:

ASTM A 953 S70003 is a type of seamless tubing made from a special steel alloy that includes chromium, nickel, and silicon. This unique combination makes it highly resistant to corrosion and capable of withstanding high temperatures, making it an ideal choice for various industrial applications. It is particularly useful in settings where equipment is exposed to harsh conditions, such as in valves and pumps used in chemical processing or oil and gas industries.

- Benefits of ASTM A 953 S70003:**
- **High Corrosion Resistance:** This tubing can endure exposure to corrosive substances, ensuring longevity and reliability in tough environments.
 - **High-Temperature Suitability:** It can function effectively in high-temperature applications, making it versatile for use in different industrial sectors.
 - **Strong and Durable:** With a minimum tensile strength of 540 MPa and good elongation properties, this tubing is both strong and flexible, allowing it to handle rigorous demands without failure.

ASTM A 954 S70003

Category: Pipe

Austenitic Chromium-Nickel-Silicon Alloy Steel Seamless and Pipe

Key Features:

- Corrosion resistance
- High-temperature stability

Technical Specifications:

Tensile Strength	540 Mpa min.
Yield Strength	240 Mpa min.
Elongation	40% min.

Use Cases:

- Used in the construction of pipelines for corrosive and high-temperature applications.

ASTM A 954 S70003 Summary:

ASTM A 954 S70003 is a type of strong and durable pipe made from a special steel alloy that contains chromium, nickel, and silicon. This pipe is designed to handle tough conditions, making it perfect for use in pipelines that transport corrosive substances or operate in high temperatures. By being seamless, it means there are no joints that can weaken the structure, ensuring a reliable performance.

Key benefits of ASTM A 954 S70003 include:

- **Corrosion Resistance:** It can withstand harsh environments without rusting or degrading.
- **High-Temperature Stability:** It maintains its strength and shape even in very hot conditions, making it ideal for challenging industrial applications.

Overall, this type of pipe is essential for industries that need reliable and long-lasting materials for their pipelines, ensuring safety and efficiency in transporting various substances.

ASTM A 968 S70003

Category: Bar and Shape

Chromium, Chromium-Nickel, and Silicon Alloy Steel Bars and Shapes

Key Features:

- Corrosion and heat resistance
- Versatile applications

Technical Specifications:

Tensile Strength	540 Mpa min.
Yield Strength	240 Mpa min.
Elongation	40% min.
Hardness	95 BHN max.

Use Cases:

- Used in manufacturing structural components for engineering applications.

ASTM A 968 S70003 Summary:

ASTM A 968 S70003 is a type of steel that is specially made with chromium, nickel, and silicon. This unique combination gives the steel excellent resistance to both corrosion and heat, making it a reliable choice for many different uses. It's particularly effective in environments where durability is crucial, such as in manufacturing and engineering.

Key benefits of ASTM A 968 S70003 include:

- **Corrosion and Heat Resistance:** *This steel can withstand harsh conditions without degrading, ensuring longevity and safety in structures.*
- **Versatile Applications:** *It can be used in a variety of engineering projects, especially for creating strong structural components that require dependable materials.*

With a minimum tensile strength of 540 Mpa, it also boasts good flexibility and hardness, making it a strong performer in demanding applications.

Manufacturing & R&D Processes

Investment Casting Process

A manufacturing process that involves creating a wax pattern, coating it with a ceramic shell, and then melting the wax to create a mold for casting metal components.

Capacity: High-volume production

Workflow / Steps:

1. Die inspection
2. Robotic shelling
3. Final machining
4. CMM inspection

Machinery Used:

- CNC Machines
- VMC Machines
- HMC Machines
- VTL Machines

Sand Casting

A versatile manufacturing technique for producing large, complex, and durable castings.

Capacity: 0 Kg Single Piece Job Handling Capacity

Workflow / Steps:

1. Pattern Making
2. Mold Preparation
3. Core Placement
4. Mold Assembly
5. Metal Pouring
6. Cooling
7. Mold Breakout
8. Fettling & Cleaning
9. Machining & Finishing

Audience & Market

Target Buyer Personas

- Manufacturing and production managers

Target Industries

- manufacturing
- technology
- construction
- energy

Value Propositions (USPs)

- Precision-engineered components with high dimensional accuracy.
- Capability to produce complex geometries with repeatable accuracy.
- Integrated manufacturing facility with advanced quality assurance processes.
- High-quality nickel and nickel alloy castings
- Compliance with international standards
- Suitable for a wide range of industrial applications
- High-quality materials with excellent mechanical properties
- Wide range of applications in various industries
- Compliance with international ASTM standards
- Precision-engineered components with high tolerance and complex geometries.
- Cost-effective production for low- to medium-volume requirements.
- Versatility in size and material flexibility.

Brand Tone

Professional and enterprise-oriented

Digital Maturity & SEO Observations

Overall Maturity Score: 71/100

Website Quality Score: 100/100

SEO Readiness Score: 75/100

SEO Gaps & Opportunities:

N/A

R&D Insights & Recommendations

- Leverage advancements in additive manufacturing to enhance the precision of investment castings, potentially reducing lead times and costs while increasing complexity in designs.
- Explore the growing demand for corrosion-resistant materials in industries such as oil and gas, where ASTM A 494 N12MV and ASTM A 511 N08904 products could be marketed as premium solutions.
- Capitalize on the rising trend of sustainability by developing eco-friendly casting processes and materials, which can attract environmentally conscious clients and comply with emerging regulations.
- Expand service offerings to include digital solutions such as predictive maintenance and real-time monitoring for clients using seamless stainless steel mechanical tubing in critical applications.

Marketing & Sales Strategy

Content Strategy

Platforms: LinkedIn, Instagram, YouTube

Post Types: Video, Carousel, Infographic, Blog Post

Core Themes:

- Behind the scenes manufacturing
- Product demos
- Customer testimonials
- Industry trends

Creative Concepts

Video - Procurement Managers

A 30s fast-paced reel showing the CNC machine cutting steel.

Highlighting precision and speed.

Carousel - Manufacturing and production managers

Showcase of various castings and their applications in different industries.

Demonstrating versatility and application.

Competitor Landscape

Competitor	Region	Threat	Differentiator
Gokul Castings	India	High	They offer a wide range of castings at competitive prices.
Siddhivinayak Castings	India	Medium	Expertise in custom castings.
Bharat Forge	India	High	Strong brand presence in the automotive sector.
Anand Engineering	India	Medium	Focus on sustainable manufacturing practices.
Kumar Metal Industries	India	Medium	Specialized in high-performance alloys.
Alcoa Corporation	Global	High	Industry leader in lightweight metals.
Precision Castparts Corp.	Global	High	Focus on aerospace and defense applications.
Thyssenkrupp AG	Global	High	Diverse portfolio including automotive and construction.
Metal Technologies, Inc.	Global	Medium	Customized casting solutions for various industries.
Castrol Limited	Global	Medium	Strong brand in lubrication and metalworking fluids.

LinkedIn Outreach Playbook

50 Targeted pitch messages to convert customers, categorized by persona.

Persona: CEO

- I noticed many manufacturers struggle with supply chain disruptions. We solved this by streamlining our production process, ensuring timely delivery and consistency.
- I've observed that companies often face challenges with quality assurance. We tackled this by implementing rigorous quality control measures that exceed industry standards.
- I see that many businesses in our sector are looking to reduce costs while maintaining quality. We provide high-quality castings that optimize production efficiency, ultimately saving costs in the long run.
- I noticed that many manufacturers find it hard to source specialized materials. We addressed this by offering a diverse range of advanced materials tailored to specific applications.
- I've learned that many companies wrestle with scaling production. We've developed scalable casting solutions that adapt to your production needs without compromising quality.

Persona: Production Manager

- I noticed that production managers frequently deal with equipment downtime. We solved this by investing in state-of-the-art technology that minimizes downtime and maximizes productivity.
- I see that maintaining tight production schedules can be challenging. We've implemented just-in-time manufacturing practices that ensure on-schedule delivery of your required castings.
- I've observed that many production teams struggle with quality inconsistencies. We offer a robust quality management system that guarantees uniform quality across all products.
- I understand that sourcing high-performance materials is often a hurdle for production managers. We provide a reliable supply of high-grade materials to keep your production lines running smoothly.
- I noticed many managers are overwhelmed by compliance regulations. We take the burden off by ensuring all our products meet or exceed the latest industry standards.

Persona: Procurement Manager

- I've seen procurement managers often face difficulties in finding reliable suppliers. We solved this by establishing long-term partnerships with trusted suppliers to ensure consistent quality.
- I recognize that many procurement managers struggle with pricing pressures. We maintain competitive pricing while ensuring the highest quality in our castings.
- I understand that managing supplier relationships can be complex. We simplify this by providing dedicated support teams to assist you throughout the procurement process.
- I've noticed that many in your role are concerned about lead times. We address this by optimizing our production schedules to meet your urgent delivery needs.
- I see that procurement managers often have to deal with multiple suppliers for different needs. We offer a one-stop solution for all your casting requirements, streamlining your procurement process.

Persona: Quality Control Manager

- I noticed that quality control managers often deal with unexpected defects. We solved this by utilizing advanced inspection technologies to catch issues before they reach your facility.
- I've observed that many quality managers grapple with compliance issues. We ensure all our products comply with rigorous industry standards, relieving you of that concern.
- I see that maintaining a consistent quality level can be a significant challenge. We've developed processes that consistently deliver high-quality castings, ensuring your reputation stays intact.
- I understand that many quality control teams are overwhelmed with documentation. We provide clear and comprehensive documentation for all our products, making compliance much easier.
- I've learned that many quality managers are looking for ways to innovate. We're committed to continuous improvement, employing the latest technologies to enhance product quality.